

REPUBLIC OF RWANDA

EASTERN PROVINCE

KIREHE DISTRICT

KIGINA SECTOR

KIREHE ADVENTIST TECHNICAL SECONDARY SCHOOL

QUALITY ASSUARENCE QUESTIONS AND ANSWER ENGLISH And

KINYARWANDA EXPLANATIONS

i. What do you understand about Quality Assurance?

QA bisobanura iki?

English: Quality Assurance (QA): is a **systematic process** to ensure that a product or service meets desired quality standards.

Kinyarwanda: QA ni uburyo bugamije kugenzura no kwemeza ko ibicuruzwa cyangwa porogaramu bikoze neza, byujuje ibisabwa kandi byizewe.

QA ni process proactive (itangirire kare) igamije gukumira amakosa mbere y'uko aba.

ii. What are the stages/processes of Quality Assurance?

Ibyiciro bya QA ni ibi bikurikira:

1. **Analyse the requirements** : Gusobanukirwa ibisabwa (functional & non-functional)
2. **Plan the Tests** : Gutegura igipimo cy'ibizakorerwa (Test Plan)
3. **Design the Tests** : Gushushanya uko test izakorwa (Test Cases, Test Scripts)
4. **Execute the Tests and Report Defects** : Gukora test no kumenyekanisha amakosa yabonetse.
5. **Re-Test and Regression Test** :Gusubiramo test kugira ngo barebe ko ibibazo byakemutse ndetse ntibyangize ibindi bice.
6. **Run Release Tests** : Gukora igenzura rya nyuma mbere yo gusohora porogaramu.

iii. What are the main types of Quality Assurance?

Ibyiciro by'ingenzi bya QA:

1. **Functional QA** (gupima imikorere)
2. **Non-functional QA** (gupima umutekano, umuvuduko, igipimo)
3. **Automated QA** (gukoresha tools)
4. **Manual QA** (gupima n'intoki)
5. **Regression QA** (kureba ko ibyo wasannye bitangiza ibindi)
6. **User Acceptance Testing (UAT)**
7. **Compliance Testing** (kugenzura ko hubahirizwa amabwiriza)

iv. Mention the methods used in Quality Assurance.

Uburyo bukoreshwa muri QA:

- Audit (kugenzura intambwe)
- Review (gusuzuma hamwe)
- Bug detection (gushakisha amakosa)
- Design inspection (gusuzuma igishushanyo)

v. List down the standards used in Quality Assurance.

Standards zikoreshwa muri QA:

- ISO 9001
- ISO/IEC 17025
- MIL-STD-810
- ISO 26262
- Six Sigma
- IEEE Standards

vi. What are the benefits of Quality Assurance?

Inyungu za QA:

- Kwirinda amakosa mbere y'uko abaho (**Prevention of Defects**)
- Kongera kunyurwa kw'abakiriya (**Customer Satisfaction**)
- Kongera imikorere no kuzigama igihe (**Efficiency**)
- Kubahiriza amabwiriza (**Compliance**)
- Gufasha iterambere ry'ibicuruzwa (**Continuous Improvement**)

vii. Identify Quality Assurance classification.

Ibyiciro bya QA:

- **Industry Specific QA:** (Software QA, Healthcare QA, Manufacturing QA, Construction QA)

- **Process vs Product QA:**
 - **Process QA** = Kureba ko inzira ikurikiza amahame
 - **Product QA** = Kureba ko igicuruzwa kijuje ibisabwa
- **Internal vs External QA:**
 - **Internal** = Igenzurwa n'ikipe y'imbere
 - **External** = Igenzurwa n'abagenzuzi bo hanze
- **Manual vs Automated QA:**
 - **Manual** = Abantu bakora igenzura
 - **Automated** = Byifashishijwe ibikoresho by'ikoranabuhanga

viii. Differentiate Quality Assurance from Quality Control.

Aspect	QA	QC
Ibisobanuro	Group of activities for preventing defects	Group of activities for detecting defects
Focus	Process-oriented	Product-oriented
Uko bikorwa	Establishing methods and audits	Testing the final product
Igihe bikorerwa	All along the development	After development
Ingero	Verification (QA)	Validation (QC)

QA = Proactive, QC = Reactive.

ix. What are tools used in Quality Assurance?

Ibikoresho bikoreshwa muri QA:

- **Test Management Tools:** JIRA, TestRail
- **Automated Testing Tools:** Selenium, JUnit/TestNG, Cypress
- **Performance Testing Tools:** JMeter, LoadRunner
- **Static Analysis Tools:** SonarQube, ESLint
- **CI/CD Tools:** Jenkins, CircleCI
- **Version Control Tools:** Git, GitHub, GitLab
- **Bug Tracking Tools:** Bugzilla, Redmine
- **Security Testing Tools:** OWASP ZAP, Burp Suite

x. Describe techniques of Quality Assurance?

Ubuhanga bwa QA:

- **Static Testing:** Code Reviews, Static Analysis (kubikora nta kuzikoresha)
- **Dynamic Testing:** Unit Testing, Integration Testing, System Testing
- **User Acceptance Testing (UAT):** Gukoresha software hamwe n'umukiriya

- **Exploratory Testing:** Gushakisha amakosa bisanzwe nta nyandiko igenderwaho
- **Regression Testing:** Gusubiramo test kugira ngo barebe ko ibihinduwe bitangiza ibindi
- **Performance Testing:** Gupima umuvuduko, umutekano, scalability
- **Security Testing:** Gushakisha intege nke
- **Compatibility Testing:** Kureba uko software ikora kuri platforms zitandukanye
- **Smoke Testing:** Gupima imikorere fatizo mbere y'ibindi
- **Automation Testing:** Gupima hifashishijwe ibikoresho by'ikoranabuhanga (Selenium, Cypress)
- **Boundary Value Analysis:** Gupima imbibi z'agaciro
- **Equivalence Partitioning:** Gutandukanya inputs mu byiciro no gupima buri cyiciro

✳ Other Important Points Not Included in the Questions:

✓ QA irimo ibijyanye na:

- **Risk Management:** Kumenya ibyago no kubishakira igisubizo
- **Feasibility Assessment:** Kureba ko umushinga ushobora gukorwa
- **Communication Plan:** Gufasha abakozi kumenya uko bazavugana
- **Legal and Compliance Consideration:** Kureba ko software yubahiriza amabwiriza

i. What do you understand by “terms of reference”?

English: Terms of Reference (ToR) is a formal document that clearly defines the **objectives, scope, roles, and responsibilities** of a project or task.

Kinyarwanda: ToR ni inyandiko yandikwamo **intego, imbibi, inshingano** n'uburyo umushinga cyangwa igikorwa kigiye gukorwamo.

ii. Give the reason why the Terms of Reference is necessary in Quality Assurance.

English: ToR is necessary because it:

- Clarifies the scope of QA activities (irakwereka imbibi z'ibikorwa).
- Defines roles and responsibilities clearly (igena uzakora iki).
- Helps align stakeholder expectations.
- Provides guidelines and standards for the QA process.

Kinyarwanda: ToR irakenewe kugirango:

- **Isobanure neza imbibi** z'ibikorwa QA izakora.
- **Igene neza inshingano** z'buri wese.
- Ifashe mu **kuganira no kumva hamwe** n'abafatanyabikorwa.
- Igaragaze **amabwiriza** agenga QA.

iii. Outline the elements of Terms of Reference.

English:

1. Background
2. Objectives
3. Scope
4. Methodology
5. Timeline
6. Deliverables
7. Resources
8. Roles and Responsibilities
9. Reporting Requirements
10. Evaluation Criteria

Kinyarwanda:

1. Aho umushinga uhera (Background)
2. Intego (Objectives)
3. Imbibiri (Scope)
4. Uburyo bizakorwa (Methodology)
5. Igihe bizakorwa (Timeline)
6. Ibikorwa (Deliverables)
7. Ibyo bikeneye (Resources)
8. Inshingano (Roles and Responsibilities)
9. Uko raporo zizakorwa (Reporting Requirements)
10. Imiterere yo gusuzuma umusaruro (Evaluation Criteria)

iv. What is the importance of Terms of Reference?

English: Importance of ToR:

- Reduces misunderstandings between stakeholders.
- Provides a roadmap and structure for the project.
- Enhances accountability by defining roles clearly.
- Enables better planning, resource allocation, and project monitoring.

Kinyarwanda: Akamaro ka ToR:

- Irinda **kutumvikana nabi** hagati y'abafatanyabikorwa.
- Itanga **icyerekezo n'imiterere** y'umushinga.
- Ituma **inshingano z'imirimu** ziba zizwi.
- Ifasha mu **gutegura neza**, guha abantu ibikorwa, no gukurikirana umusaruro.

v. What is test case in terms of reference?

English: A test case is a set of conditions or inputs used to verify that a specific feature or requirement of the software works as intended.

It includes:

- Test case name
- Inputs required
- Expected results

Kinyarwanda: Test case ni igice kigaragaza:

- Izina rya test
- Ibizakenerwa ngo ikorwe
- Ibisubizo bitegerejwe

vi. Describe document analysis report in terms of reference.

English: A Document Analysis Report is a review of the ToR and related documents to:

- **Confirm that objectives are clear.**
- **Evaluate the completeness and consistency of requirements.**
- **Identify gaps, constraints, and assumptions.**
- **Assess feasibility and potential risks.**
- **Recommend improvements.**

Kinyarwanda: Raporo y'isesengura ry'inyandiko igamije:

- **Kureba ko intego zisobanutse.**
- **Kureba ko ibisabwa byuzuye kandi bihamye.**
- **Kugaragaza aho hari icyuho, imbogamizi, cyangwa ibisabwa bikenewe.**
- **Gusuzuma uko umushinga ushobora gushyirwa mu bikorwa.**
- **Gutanga inama zo kunoza umushinga.**

✳ Additional Important Points about ToR from the Manual

- ✓ **ToR itanga icyerekezo cya QA (guideline for QA).**
- ✓ **Igira uruhare mu kurinda amakosa no gutuma umushinga ukorwa mu rwego rw'ubuziranenge.**
- ✓ **Igira uruhare mu kugenzura ibikorwa no gutanga ishusho y'ahagomba kwibandwa.**

i. What are the components of a requirement specification document?

English:

A requirement specification document includes:

1. Title Page

2. Table of Contents
3. Executive Summary or Abstract(Incamake)
4. Introduction
5. Quality Assurance Objectives
6. Scope of Work
7. Roles and Responsibilities
8. QA Methodology and Approach
9. Test Plan
10. Test Cases and Procedures
11. Test Data
12. Risk Management
13. Quality Metrics and Reporting
14. Compliance and Standards
15. Change Management
16. Documentation and Record-Keeping
17. Approval and Sign-Off
18. Glossary
19. Appendices

Kinyarwanda:

Ibice bigize inyandiko y'ibisabwa:

1. Umutwe w'inyandiko
2. Imbonerahamwe y'ibirimo
3. Incamake
4. Iriburiro
5. Intego za QA
6. Urutonde rw'imirimo
7. Inshingano
8. Uburyo QA izakorwa
9. Gahunda y'ibizamini
10. Ibisabwa ku bizamini
11. Amakuru y'ibizamini
12. Gucunga ibibazo
13. Ibipimo by'ubuziranenge
14. Amabwiriza mpuzamahanga
15. Guhindura imikorere
16. Inyandiko zibikwa
17. Kwishyiriraho umukono
18. Inkoranyamagambo
19. Inyandiko z'inyongera

ii. Describe functional and non-functional requirements for stakeholder.

✓ **English:**

- **Functional requirements:** What the system must do (user needs), e.g., *user can log in, user can register, user can make payments.*
- **Non-functional requirements:** Constraints on how the system operates, e.g., performance, security, reliability, scalability.

✓ **Kinyarwanda:**

- **Ibisabwa bikora (functional):** Ibyo porogaramu igomba gukora (ibyo umukiriya asaba), urugero: *Kwinjira muri sisitemu, Kwiyaandikisha, Kwishura.*
- **Ibisabwa bidakora (non-functional):** Imiterere cyangwa uburyo porogaramu igomba gukora, urugero: umuvuduko, umutekano, uko igenda ikura (scalability), kwizerwa.

iii. Differentiate consistency and completeness requirement specification.

✓ **English:**

Aspect	Consistency	Completeness
Definition	All requirements use the same terms and style, no contradictions.	All required requirements are included with no missing details.
Focus	Making sure wording, terminology, and structure are aligned.	Making sure every requirement (functional/non-functional) is stated.
Importance	Avoid misunderstanding due to contradicting information.	Avoid leaving any requirement out that might cause project failure.
✓ Kinyarwanda:	Consistency	Completeness
Ingingo	Consistency (Guhuza)	Completeness (Kuzuza)
Igisobanuro	Imikoreshereze y'amagambo, imiterere no guhuza amakuru.	Kureba ko ibisabwa byose (functional na non-functional) byanditswe.
Icyibandwaho	Ukutavuguruzanya kw'amagambo n'ibisobanuro.	Kureba ko nta gisabwa cyibagiwe.
Akamaro	Kwirinda urujijo mu bisabwa.	Kwizera ko igicuruzwa gishobora gukora ibyo cyateguriwe.

iv. Identify dependencies and interactions in requirement specification.

✓ **English:**

Dependencies:

- **Certain requirements must be implemented in order or depend on other requirements.**
 - **Example:** Login must be implemented before user can access dashboards.

Interactions:

- Describes how requirements affect one another:
 - **Positive Interaction:** One requirement supports or improves another.
 - **Negative Interaction:** One requirement may conflict with another.
 - **Neutral Interaction:** No significant effect between requirements.

✓ Techniques for identifying dependencies & interactions:

- Dependency Mapping (diagramming)
- Impact Analysis
- Risk Assessment
- Traceability Matrix (connect requirements, design, and test)

✓ **Kinyarwanda:**

Ibikenerwa bigenderanye (Dependencies):

- Ibisabwa bimwe bishingira ku bindi.
 - *Urugero:* Kwiwandikisha (registration) bigomba kubanza gukorwa mbere yo kwinjira (login).

Uko bisabwa bigirana isano (Interactions):

- Ibisabwa bishobora gufashanya, guhangana, cyangwa ntibigire aho bihurira.
 - *Positive Interaction:* Ikintu kimwe gitera ikindi gukora neza.
 - *Negative Interaction:* Ikintu kimwe gishobora kubuza ikindi gukora.
 - *Neutral Interaction:* Nta sano ifatika.

✓ Techniques zo kumenya isano n'ibigenderanye:

- Gushushanya ikarita y'ibigenderanye (Dependency Mapping)
- Gusesengura ingaruka (Impact Analysis)
- Gusuzuma ibyago (Risk Assessment)
- Traceability Matrix (Guhuza ibisabwa, igishushanyo, n'ibizamini)

✳ Additional Important Point from the Manual

► Why this is critical for QA?

- To ensure no requirement contradicts another (avoids errors).
- To confirm every requirement has been accounted for (no missing features).
- To assess how changes to one requirement affect others (impact and risk).

i. What is an Inception Report?

✓ **English:** An inception report is an **initial project document** prepared at the very start of a project. It **outlines the approach, scope, objectives, and planning details** required for guiding the project implementation.

✓ **Kinyarwanda:** Inception Report ni **inyandiko ibanziriza umushinga**, igaragaza **uburyo umushinga uzakorwa, intego, imbibi (scope), n'ibindi bisobanuro** bigamije kuyobora igikorwa cy'umushinga.

ii. What is Document Review in an Inception Report?

✓ **English:** A review of all available documents related to the project (such as policies, prior studies, data) to:

- Check alignment with the project objectives.
- Identify gaps or areas that need more clarification or further exploration.

✓ **Kinyarwanda:** Gusuzuma inyandiko zose **zinyanye n'umushinga** (amabwiriza, raporo za kera, imibare) kugira ngo:

- Harebwe **uko bihura n'intego** z'umushinga.
- Hamenyekane **ahakenewe gusobanuka** cyangwa gukeneye ubushakashatsi.

iii. What do you understand by “Stakeholder Expectations”?

✓ **English:** Stakeholder expectations are the **needs, priorities, and concerns** of everyone involved in or impacted by the project including **clients, beneficiaries, sponsors, and partners**.

✓ **Kinyarwanda:** Ibyifuzo by'Abafatanyabikorwa ni **ibyo bakeneye, ibyo bashyira imbere, n'ibibazo batekereza** ku biyanye n'umushinga **harimo abakiriya, abafatanyabikorwa, abaterankunga, n'abazungukira**.

iv. What is Clarity and Completeness in the Inception Report?

✓ **English:**

- **Clarity:** The objectives, deliverables, roles, and milestones are stated clearly and precisely.
- **Completeness:** All necessary details are included (scope, stakeholders, timeline), leaving no room for ambiguity.

✓ **Kinyarwanda:**

- **Gusobanuka (Clarity):** Intego, ibyitezwe, inshingano, n'amatariki bigomba kugarazwa mu buryo busobanutse.

- **Kuzuza (Completeness):** Amakuru yose asabwa (imbibi z’umushinga, abafatanyabikorwa, ingengabihe) agomba kuba arimo, ntakintu cyibagiranywe.

v. What is the Feasibility and Risks?

✓ English:

Feasibility is assessing if the project can be realistically implemented within the available resources, time, and constraints.

Risks are potential problems or threats that might affect the project’s success, such as budget shortages or delays.

✓ Kinyarwanda:

Feasibility ni ukureba **niba umushinga ushobora gukorwa** ukurikije igihe, amikoro, n’ibindi bisabwa.

Risks ni **ibibazo cyangwa imbogamizi** zishobora kubangamira umushinga, nko kubura amafaranga, gutinda, n’ibindi.

vi. Describe Document Assumptions and Constraints.

✓ English:

- **Assumptions:** Things we consider true for planning the project, such as availability of **staff, cooperation from stakeholders**, or **stable technology**.
- **Constraints:** Limitation factors like budget restrictions, regulations, or tight deadlines that might affect project execution.

✓ Kinyarwanda:

- **Ibidufasha kwemeza ko bizagenda neza (Assumptions):** Nko kumenya ko abakozi bazaboneka, ko abafatanyabikorwa bazafatanywa, cyangwa ko ikoranabuhanga rizakora neza.
- **Inzitizi (Constraints):** Imbogamizi nk’amafaranga make, amabwiriza akomeye, cyangwa igihe gito cyatanzwe.

vii. Describe Document Analysis Findings.

✓ **English:** These are the **key insights and results** gained from the review of the project’s documents:

- **Identified gaps and inconsistencies.**
- **Areas requiring more information.**
- **Stakeholder priorities and constraints.**
- **Risk areas and proposed mitigation.**
- **Final recommendations for moving forward.**

✓ **Kinyarwanda:**

Ni ibyavuye mu isesengura ry'inyandiko z'umushinga:

- Ahantu hari icyuho cyangwa ibitanoheranye.
- Ahakenewe amakuru y'inyongera.
- Ibyifuzo n'imbogamizi z'abafatanyabikorwa.
- Ahagaragaye ingorane n'uko zakemurwa.
- Inama z'uburyo umushinga ukwiye gukomeza.

✳ **Additional Important Points for Inception Report:**

✓ **The Inception Report** is crucial for aligning stakeholders, confirming feasibility, identifying constraints, and making sure **all parties have a common understanding** of the project.

✓ **Techniques used:**

- Document review, stakeholder consultations, risk assessment, feasibility studies.

✓ **Output:**

- Provides a **baseline for project execution**, making sure the project is viable and aligned with objectives.

i. What do you understand by “Test Plan”?

✓ **English:** A **Test Plan** is a **detailed document** that describes the testing objectives, scope, resources, approach, schedule, and deliverables for a testing activity.

It guides the QA team through the testing process.

✓ **Kinyarwanda:** Test Plan ni **inyandiko irambuye** igaragaza intego, imbibi, ibikoresho, uburyo, igihe, n'ibizava mu gikorwa cyo gupima.

ii. What are types of Test Plan?

✓ **English:**

- **Master Test Plan** (Overall testing approach for the project)
- **Unit Test Plan** (Tests for individual units/modules)
- **Integration Test Plan** (Tests for combined units)
- **System Test Plan** (Tests for the complete system)
- **Acceptance Test Plan** (Tests for user acceptance)

✓ **Kinyarwanda:**

- Master Test Plan: Gahunda rusange yo gupima umushinga wose
- Unit Test Plan: Gahunda yo gupima igice kimwe cya porogaramu
- Integration Test Plan: Gahunda yo gupima ibice bihujwe
- System Test Plan: Gahunda yo gupima porogaramu yose
- Acceptance Test Plan: Gahunda yo gupima ibyemeza n'umukiriya

iii. What are the components of a Test Plan?

✓ English:

- Introduction
- Objectives
- Scope
- Test Items
- Features to be Tested
- Features Not to be Tested
- Testing Approach
- Resources and Responsibilities
- Test Schedule
- Risk and Contingency
- Entry and Exit Criteria
- Deliverables

✓ Kinyarwanda:

- Iriburiro
- Intego
- Imbibizi z'ibizakorwa
- Ibizapimwa
- Ibizakorwa test
- Ibizazagerazwa
- Uburyo bizakorwa
- Abantu n'ibikoresho
- Ingengabihe y'ibizamini
- Ingaruka zishobora kubaho
- Ibisabwa mbere no nyuma y'ibizami
- Ibisohoka

iv. Mention Tools Used for Testing the System.

✓ English:

- **Automated Testing Tools:** Selenium, Cypress, JUnit, TestNG
- **Test Management Tools:** JIRA, TestRail, Xray

- **Bug Tracking Tools:** Bugzilla, Redmine, Mantis
- **Performance Testing Tools:** JMeter, LoadRunner
- **Static Analysis Tools:** SonarQube, ESLint

✓ **Kinyarwanda:**

- Automated Testing Tools: Selenium, Cypress, JUnit, TestNG
- Test Management Tools: JIRA, TestRail, Xray
- Bug Tracking Tools: Bugzilla, Redmine, Mantis
- Performance Testing Tools: JMeter, LoadRunner
- Static Analysis Tools: SonarQube, ESLint

v. Identify Test Criteria to Consider During Testing System.

✓ **English:**

- **Entry Criteria:** What must be met before starting testing (e.g., requirements available, test data prepared).
- **Exit Criteria:** What must be met to end testing (e.g., no critical defects open, test coverage achieved).
- **Acceptance Criteria:** What the system must achieve for stakeholders to accept it.

✓ **Kinyarwanda:**

- **Ingingo zo gutangira:** Ibisabwa kugira ngo igipimo gitangire (ibisabwa, amakuru y'ibizami)
- **Ingingo zo gusoza:** Ibisabwa kugira ngo igipimo kirangire (nta makosa akomeye asigaye)
- **Ingingo zo kwemera:** Ibisabwa kugira ngo umukiriya yemere ko igicuruzwa gishobora gukoreshwa

vi. Identify Test Techniques Used for Testing the System.

✓ **English:**

- **Black Box Testing**
- **White Box Testing**
- **Unit Testing**
- **Integration Testing**
- **System Testing**
- **User Acceptance Testing (UAT)**
- **Regression Testing**
- **Boundary Value Analysis**
- **Exploratory Testing**

vii. Describe Test Cases in Testing System.

✓ **English:** A Test Case is a set of conditions, inputs, and expected outcomes used to verify that a feature or requirement works as intended.

✓ **Kinyarwanda:** Test Case ni urutonde rw'ibisabwa, amakuru, n'ibizava mu gikorwa cy'igipimo, bigamije kwemeza ko igice cya porogaramu gikora neza.

viii. What is Test Data in System Testing?

✓ **English:** Test data is the set of inputs used for verifying the behavior of a software system during test execution.

✓ **Kinyarwanda:** Test Data ni amakuru ashyirwa muri porogaramu kugira ngo igipimo kigaragaze uko igikorwa.

ix. Describe Types of Test Data Used in Testing System.

✓ **English:**

- Valid Test Data
- Invalid Test Data
- Boundary Test Data
- Large Volume Test Data
- Special Characters or Malformed Inputs
- Null or Empty Inputs

✓ **Kinyarwanda:**

- Amakuru meza (Valid Test Data)
- Amakuru mabi (Invalid Test Data)
- Amakuru y'umupaka (Boundary Test Data)
- Amakuru menshi (Bulk/Volume Test Data)
- Amakuru adasanzwe cyangwa afite ibimenyetso bidasanzwe
- Amakuru adahari (Null or Empty Inputs)

x. Give the Importance of Test Data.

✓ **English:**

- Enables verification of requirements
- Helps in identifying defects and gaps
- Supports automated and manual testing
- Ensures test coverage and reliability
- Enables reproduction of defects and results

✓ **Kinyarwanda:**

- Ifasha kwemeza ko ibisabwa byubahirijwe
- Ifasha kubona amakosa n'ibyuho
- Ifasha igipimo cya porogaramu (automated na manual)
- Ituma igipimo gishingira ku makuru nyakuri
- Ituma amakosa ashobora gusanwa no gusubiramo igipimo

xi. Describe Test Execution Scripts.

✓ **English:** Test Execution Scripts are **automated or manual instructions** that describe the steps, inputs, and expected results for executing test cases.

✓ **Kinyarwanda:** Test Execution Scripts ni **inyandiko zirimo amabwiriza** akoreshwa mu gupima igice cya porogaramu, harimo intambwe, amakuru, n'ibizava mu gikorwa.

xii. What are the Components of Test Delivery Plan?

✓ **English:**

- Testing Objectives
- Testing Scope
- Testing Timeline/Schedule
- Resource and Role Assignments
- Entry and Exit Criteria
- Risk Management and Contingency Plans
- Test Deliverables
- Test Environment Specifications
- Communication and Reporting Plans

✓ **Kinyarwanda:**

- Intego zo gupima
- Imbibizi zo gupima
- Ingengabihe y'ibizamini
- Inshingano n'abakozi
- Ibisabwa mbere na nyuma yo gupima
- Imicungire y'ibibazo no gutegura igisubizo
- Ibisohoka mu gupima
- Imiterere y'ibidukikije byo gupima
- Itumanaho no gutanga raporo

✳ Additional Important Points from the Manual:

✓ **The Test Plan** guarantees **clear communication, well-prepared test data**, and a **roadmap** for achieving quality in software projects.

✓ **A well-prepared Test Plan** improves efficiency and ensures that all stakeholders understand their roles and responsibilities.

i. What are the Key Components of a Testing Environment?

✓ English:

- **Hardware** (Servers, Computers, Devices)
- **Software** (OS, Test Tools, Applications)
- **Network** (Connectivity, Bandwidth, Firewall)
- **Database** (Test databases, data storage)
- **Test Scripts & Test Cases**
- **Documentation** (Requirements, Test Plan)

✓ Kinyarwanda:

- **Ibyuma (Hardware):** Seriveri, mudasobwa, ibikoresho bitandukanye.
- **Porogaramu (Software):** Sisitemu y'imikorere, ibikoresho byo gupima, porogaramu.
- **Umuyoboro (Network):** Ihuriro rya internet, umuvuduko, firewall.
- **Base y'Amakuru:** Aho amakuru y'ibizami abikwa.
- **Scripts & Cases z'ibizamini.**
- **Inyandiko:** Ibisabwa, Gahunda y'ibizami.

ii. Explain the Hardware Infrastructure used in Testing Environment.

i. Describe the review test plan.

✓ English:

Reviewing the test plan means **evaluating its completeness, correctness, and alignment** with requirements before test execution.

This involves verifying test objectives, scope, resources, schedules, and risk identification.

✓ Kinyarwanda:

Gusuzuma Test Plan bisobanura **kureba ko irimo ibyo ikwiye, ko idafite amakosa, no kumenya ko ihuye n'ibisabwa.**

Bikubiyemo gusuzuma intego, imbibi, amikoro, ingengabihe, no kumenya ingorane zishobora kubaho.

ii. What are key aspects to select test cases?

✓ English:

- **Requirement coverage** (igipimo cy'ibisabwa)
- **Risk priority** (ibifite ingaruka nyinshi)
- **Business criticality** (ibizami bijyanye n'imikorere y'ingenzi)

- Reusability (ibizami byakoreshwa kenshi)
- Test data availability (amakuru yo gupima aboneka)
- Frequency of feature usage (igice gikoreshwa kenshi)

iii. Describe test execution.

✓ English:

Test Execution is the process of running test cases, recording results, and verifying the software behavior against requirements.

✓ Kinyarwanda:

Gukora igipimo (Test Execution) ni igikorwa cyo gukoresha test cases, kwandika ibisubizo, no kugenzura ko porogaramu ikora uko bikwiye.

iv. Explain record defects.

✓ English:

Recording defects means **logging issues** discovered during testing, including details like description, steps to reproduce, severity, and status.

✓ Kinyarwanda:

Kwandika inenge bisobanura **kuzuza amakuru y'ikibazo cyabonetse**: ibisobanuro, intambwe zo kukibona, uburemere, hamwe n'uko kimeze.

v. What is the Purpose of Recording Defects?

✓ English:

- To track and manage errors for timely resolution
- To improve software quality
- To evaluate developer performance and root causes

✓ Kinyarwanda:

- Gukurikirana no gukemura amakosa vuba
- Kongera ubwiza bwa porogaramu
- Gusuzuma imikorere y'abakora kode no kumenya imizi y'ikibazo

vi. Outline Key Steps to Record Defects.

✓ English:

1. Identify the defect.
2. Reproduce it reliably.
3. Record description, steps, and evidence.
4. Assign severity and priority.
5. Log in defect tracking tool.

✓ Kinyarwanda:

6. Menya ikosa.

7. Garagaza uko rishobora kugaruka.
8. Andika ibisobanuro, intambwe n'ibimenyetso.
9. Shyiraho uburemere n'icyihutirwa.
10. Bika ikosa muri porogaramu igenzura amakosa.

vii. What are Best Practices for Recording Defects?

✓ English:

- Be concise and precise.
- Include screenshots or evidence.
- Clearly state expected vs. actual results.
- Assign correct severity and priority.
- Avoid duplicate defect entries.

✓ Kinyarwanda:

- Kwandika mu buryo busobanutse.
- Gutanga ibimenyetso (amashusho).
- Kugaragaza igisubizo cyari giteganyijwe n'icyabonetse.
- Gutanga urwego rw'uburemere n'icyihutirwa nyacyo.
- Kwirinda kwandika amakosa asubiwemo.

viii. What is Monitoring Test Progress?

✓ English:

Monitoring test progress: means tracking test activities, status, and results to assess testing efficiency and determine if the project is on track.

✓ Kinyarwanda:

Gukurikirana aho igipimo kigeze bisobanura kureba uko igipimo kirimo gukorwa, ibisubizo byacyo, no kumenya niba umushinga ukomeje neza.

ix. What is the Purpose of Monitoring Test Progress?

✓ English:

- To ensure test objectives are met.
- To detect delays or issues early.
- To measure test coverage and efficiency.
- To enable informed decisions.

✓ Kinyarwanda:

- Kugenzura ko intego z'ibizamini zigezweho.
- Kumenya ibibazo cyangwa gutinda hakiri kare.
- Gupima igipimo cy'ibizamini no kumenya uko bikorwa.
- Gufasha gufata ibyemezo bishingiye ku makuru.

x. Mention Key Components of Monitoring Test Progress.

✓ English:

- Test execution status.
- Defect status and trends.
- Test coverage metrics.
- Resource utilization.
- Risk and issue status.

✓ Kinyarwanda:

- Aho igipimo kigeze.
- Aho amakosa ageze n’uko agenda.
- Igipimo cy’ibipimwa.
- Imikoreshereze y’abakozi.
- Imiterere y’ibibazo n’ingaruka.

xi. What are Best Practices for Recording Defects? (Same as above)

✓ See vii. What are Best Practices for Recording Defects?

xii. Explain Document Test Result.

✓ English:

Documentation of test results means **recording test outcomes**, including **test case status**, **defects found**, **evidence**, and **evaluation against acceptance criteria**.

✓ Kinyarwanda:

Kwandika ibisubizo by’ibizamini bisobanura **kubika uko igipimo cyagenze**, amakosa yabonetse, ibimenyetso byayo, no kureba niba bihura n’ibyo umukiriya yasabye.

✳ Additional Points from the Manual:

✓ Helps build a historical record for future reference and audits.

✓ Enables continuous improvement by analyzing test results and defect patterns.

i. Describe test data collection in software development.

✓ English:

Test data collection is the process of **gathering inputs, parameters, and sample data** required for test execution.

It ensures that test cases can be run accurately and reliably, simulating real-life scenarios.

✓ **Kinyarwanda:**

Gukusanya amakuru y'ibizami ni igikorwa cyo **kwegeranya ibipimo, amakuru y'ibanze, n'ibigenderwaho** kugira ngo igipimo gishobore gukorwa neza kandi gitange ibisubizo byizewe.

ii. Talk about Test Artifacts in software development.

✓ **English:**

Test Artifacts are documents and files created and used during the testing process.

Examples:

- Test Plans
- Test Cases
- Test Scripts
- Test Data
- Test Reports
- Defect Logs

These artifacts guide the QA process and provide evidence for review and audits.

✓ **Kinyarwanda:**

Ibikoresho by'Ibizamini (Test Artifacts) ni inyandiko n'amadosiye akoreshwa mu gikorwa cyo gupima.

Urugero:

- Test Plan (Gahunda y'ibizami)
- Test Cases (Ibisabwa gupimwa)
- Test Scripts (Amabwiriza y'ibizami)
- Test Data (Amakuru y'ibizami)
- Test Reports (Raporo y'ibizami)
- Defect Logs (Urutonde rw'amakosa)

iii. Describe test execution data.

✓ **English:**

Test execution data is the information generated when test cases are run. It includes:

- Inputs used
- Results observed
- Status (Pass, Fail, Blocked)
- Execution timestamps and environment information

✓ **Kinyarwanda:**

Amakuru y'imigendekere y'ibizami ni amakuru aboneka iyo igipimo gikoreshwa. Arimo:

- Amakuru yinjiye (Inputs)
- Ibisubizo byabonetse

- Imiterere y'igipimo (Yatsinze, Yatsinzwe, Yahagaritswe)
- Igihe igipimo cyakozwe ndetse n'imiterere y'ibidukikije

iv. Explain Test Logs in test results.

✓ English:

Test Logs are detailed records of test execution. They capture:

- Test case ID and description
- Execution results
- Errors and messages observed
- Tester comments and timestamps

✓ Kinyarwanda:

Test Logs ni inyandiko irambuye igaragaza igipimo cyakozwe. Irimo:

- Indangamibare y'igipimo n'ibisobanuro byacyo
- Ibisubizo byagaragaye
- Amakosa cyangwa ubutumwa bugaragaye
- Ibitekerezo by'uwayikoze n'igihe igipimo cyakozwe

v. Explain Test Findings in test results.

✓ English:

Test Findings summarize the **outcomes and observations** from testing. They highlight:

- What worked well
- What failed or needs improvement
- Areas requiring further investigation
- Potential impact on quality or project timeline

✓ Kinyarwanda:

Test Findings ni ibivuye mu gikorwa cyo gupima, bikubiyemo:

- Ibikorwa byagenze neza
- Amakosa cyangwa ibikwiye kunozwa
- Ahakenewe gukorerwa irindi genzura
- Ingaruka kuri gahunda y'umushinga cyangwa ubuziranenge

vi. What is Defects in test results?

✓ English:

A **Defect** is any **discrepancy between expected and actual results**. It is an error, bug, or issue identified during testing.

✓ **Kinyarwanda:**

Defect ni ikosa cyangwa ikibazo kigaragara iyo igisubizo cyari giteganyijwe gitandukanye n'icyabonetse.

vii. Define Test Metrics in test results.

✓ **English:**

Test Metrics are **measures used to assess testing performance and effectiveness**. Examples:

- **Number of test cases executed**
- **Percentage of test cases passed or failed**
- **Defect density**
- **Test coverage percentage**
- **Mean time to detect and fix defects**

✓ **Kinyarwanda:**

Test Metrics ni **ibipimo bikoreshwa mu gusuzuma imikorere y'igipimo**. Ingero:

- **Umubare w'ibizami byakozwe**
- **Ijanisha ry'ibizami byatsinze cyangwa byatsinzwe**
- **Ubucucike bw'amakosa (defect density)**
- **Ijanisha ry'ibipimo byakozwe (test coverage)**
- **Igihe gikoreshwa mu kumenya no gukosora amakosa**

✳ **Additional Important Points from the Manual:**

- Test results guide **project decision-making** (Kwereka uko umushinga uhagaze).
- Good recording of test results allows for **audit trails** and improves future test planning.

i. What is the User Acceptance Testing (UAT)?

✓ **English:**

UAT is the final phase of software testing where the **end-users or stakeholders** test the system to ensure it meets their requirements and is ready for deployment.

✓ **Kinyarwanda:**

UAT ni igice cya nyuma cy'igerageza rya porogaramu, aho **abakoresha cyangwa abafatanyabikorwa** bapima system kugira ngo bemeze ko yujuje ibyo bakeneye kandi ishobora gukoreshwa.

ii. What are the Characteristics of UAT?

✓ **English:**

- Done by actual end-users.
- Focuses on business requirements.
- Evaluates readiness for deployment.
- Uses real-world scenarios and data.
- Final quality check before release.

✓ Kinyarwanda:

- Ikorwa n'abakoresha nyakuri.
- Irebana n'ibisabwa mu bucuruzi.
- Igenzura niba porogaramu yiteguye gukoreshwa.
- Ikoresha amakuru n'imiterere yo mu buzima busanzwe.
- Ni igenzura rya nyuma mbere yo gusohora porogaramu.

iii. What are the Objectives of UAT?

✓ English:

- Ensure the system meets end-users' needs.
- Validate all business requirements.
- Identify critical defects before release.
- Confirm the software is user-friendly.
- Minimize post-release issues.

✓ Kinyarwanda:

- Kwemeza ko system yujuje ibyo umukiriya akeneye.
- Kugenzura ko ibisabwa byose byujujwe.
- Kugaragaza amakosa akomeye mbere yo gutanga.
- Kwemeza ko porogaramu yorohereza abakoresha.
- Kugabanya ibibazo nyuma yo gushyira ku isoko.

iv. Discuss the UAT Process.

✓ English:

1. **Plan UAT** – Define objectives, test scenarios, and acceptance criteria.
2. **Design Test Cases** – Based on user requirements and business scenarios.
3. **Prepare the Test Environment** – Similar to production.
4. **Execute Test Cases** – End-users run the test.
5. **Log Defects** – Record any issues found.
6. **Evaluate Results** – Assess if acceptance criteria are met.
7. **Sign Off** – Final agreement to launch.

✓ Kinyarwanda:

1. **Gutunganya UAT** – Kugaragaza intego, ibizami, n'ibigenderwaho.
2. **Gushushanya Test Cases** – Gushingira ku bisabwa n'ibikorwa by'ubucuruzi.
3. **Kugira Ahantu ho Gupima** – Hakoresha ibidukikije bisa n'ibizakoresha.
4. **Gukora Ibizami** – Abakoresha nyakuri bakora igipimo.
5. **Kwandika Amakosa** – Gutara ibibazo byagaragaye.
6. **Gusuzuma Ibisubizo** – Kureba ko ibisabwa byagezweho.
7. **Gutanga icyemezo** – Kwemeza ko porogaramu ishobora gukoreshwa.

v. What are the Benefits of UAT?

✓ **English:**

- Ensures quality and readiness for deployment.
- Builds user trust and satisfaction.
- Reduces risk of post-production defects.
- Confirms the software meets actual needs.
- Enables stakeholders' sign-off.

✓ **Kinyarwanda:**

- Yemeza ko porogaramu igezweho kandi imeze neza.
- Yongera icyizere no kunyurwa kw'abakoresha.
- Igabanya ingaruka z'amakosa nyuma yo gutanga.
- Yemeza ko porogaramu yujuje ibyo abakiriya bakeneye.
- Ifasha abafatanyabikorwa kwemeza ko igikorwa cyarangiye.

vi. What are the Types of UAT?

✓ **English:**

- **Alpha Testing:** Done internally before external review.
- **Beta Testing:** Done by actual end-users in a real environment.
- **Contract Acceptance Testing:** According to agreement between parties.
- **Regulation Acceptance Testing:** To confirm compliance with regulations.
- **Operational Acceptance Testing:** Ensures software operates smoothly in its environment.

✓ **Kinyarwanda:**

- **Alpha Testing:** Ikorera imbere mu kigo mbere y'uko itangwa hanze.
- **Beta Testing:** Ikorera n'abakoresha nyakuri ahantu hasanzwe.
- **Contract Acceptance Testing:** Igamije kwemeza ko amasezerano yubahirijwe.
- **Regulation Acceptance Testing:** Igamije kwemeza ko porogaramu yubahiriza amabwiriza.

- **Operational Acceptance Testing:** Igamiye kwemeza ko porogaramu ikora neza mu bidukikije.

vii. What is Usability Testing?

✓ English:

Usability Testing is an evaluation method to **assess how easily and effectively end-users can interact** with the software, focusing on ease of use, efficiency, and user satisfaction.

✓ Kinyarwanda:

Usability Testing ni igipimo kigamiye **kureba ukuntu abakoresha bashobora gukoresha porogaramu byoroshye kandi bikagenda neza**, hibandwa ku koroshya imikoreshereze, umuvuduko, no kunyurwa kw'abakoresha.

viii. What is a Security Vulnerability?

✓ English:

A security vulnerability is a **weakness or flaw** in a system that can be exploited by threats (such as hackers), **potentially leading to unauthorized access, data loss, or damage**.

✓ Kinyarwanda:

Inenge mu mutekano ni **intege nke cyangwa ikosa** rishobora gukoreshwa n'abantu bafite umugambi mubi (nk'abajura) bikavamo **kwinjira mu buryo butemewe, gutakaza amakuru**, cyangwa **kwangiza system**.

★ Additional Point from the GENQA501 Manual:

✓ UAT is a critical gate for quality.

It serves as the **final quality checkpoint** that ensures software meets both **user expectations** and **business needs** before going live.

✓ i. What is a Recommendation Report?

Recommendation Report: ni inyandiko y'ibitekerezo n'ibisubizo byatanze hashingiwe ku isesengura ryakozwe nyuma yo gukora software testing.

It's a **formal document** that summarizes findings from the testing phase and provides **suggested actions** to improve or make decisions about a software system.

🔍 Ibarimo:

- **Ibyabonetse mu bizamini (findings)**
- **Ibibazo byagaragaye**
- **Inama (recommendations) ku buryo software yakosorwa cyangwa yanozwa**
- **Ingaruka zo kutagira icyo gihinduka**

✓ ii. What are the components of a Recommendation Report?

□ **Ibigize recommendation report** (according to the manual) ni ibi bikurikira:

1. **Title Page** – Uru rupapuro rurimo:
 - Izina ry’inyandiko
 - Abo yateguwe na bo
 - Itariki
2. **Table of Contents** – Imbonerahamwe y’ibikubiye mu nyandiko
3. **Executive Summary** – Incamake y’ibyingenzi
 - Findings, key risks, inama z’ingenzi
4. **Introduction** – Intangiriro
 - Icyatumye report ikorwa, intego zayo
5. **Methodology** – Uburyo hakoze assessment
 - Test plans, testing techniques, etc.
6. **Findings Summary** – Ibyabonetse mu igenzura
 - Defects, inconsistencies, bugs...
7. **Recommendations** – Inama zifatika
 - Icyo wakora ku kibazo runaka
8. **Justification** – Impamvu z’ayo mahitamo
 - Ni gute iyi recommendation izafasha
9. **Benefits and Impacts** – Inyungu n’ingaruka
 - Zizaturuka ku gushyira mu bikorwa recommendations
10. **Conclusion** – Umwanzuro
11. **Appendices** – Inyongera (test cases, charts, logs...)

✓ iii. What are the Benefits and Impacts of a Recommendation Report?

📁 **Inyungu (Benefits):**

- □ It helps **decision-makers** take informed actions.
- 📋 It ensures that **issues found during QA** are addressed.
- ✓ Igabanya ibyago byo gukora software idahuye n’ibyifuzo.
- 💡 It leads to **continuous improvement** of the system.
- □ Irakoreshe nk’inyandiko ishingirwaho mu cyemezo cya final approval.

⚠ **Impacts (Ingaruka):**

- 📉 Iyo utayikoze, project irashobora kugenda nabi cyangwa idasohoka.
- 💰 It saves time and money by preventing **recurring defects**.

-  Igira uruhare mu gutuma system iba reliable kandi efficient.

✓ iv. Describe the Archive Test Artifacts

Test artifacts ni inyandiko cyangwa ibimenyetso biba byarakozwe igihe QA yakozwe (during software testing lifecycle).

 **Archive** means to **store these test artifacts** in an organized, retrievable manner for future use.

Examples of test artifacts to be archived:

1. ✓ **Test Plans** – Document describing scope, approach, resources, and schedule of testing
2. □ **Test Cases** – Instructions to test a particular functionality
3. □ **Test Reports** – Summary of results from test executions
4.  **Defect Logs** – List of bugs found during testing
5.  **Metrics & KPIs** – Testing statistics (like pass/fail %, bug density)
6.  **User Acceptance Testing (UAT) Results**
7.  **Communication logs** – Emails, feedback, clarifications

Why archive test artifacts?

- **For compliance and auditing**
- **Kugira ibimenyetso byo gusubiraho testing process**
- **Gufasha QA teams mu bindi bihe (future releases)**
- **Guhuza amakuru hagati y'abakozi bashya n'abasanze**

BEST WISHES TO YOU ALL !!!!!!!!!!!!!!!